

MCA01 – Due: Fri., Oct. 02, 2026, before 11:50 PM

**Electrical and Computer Engineering Department
University of Puerto Rico at Mayagüez**



**COMMUNICATIONS THEORY I
MATLAB CLASS ASSIGNMENT 02 (MCA02)
Modeling a Double Sideband Suppressed Carrier
Communications System for Underwater Analog
and Digital Acoustic Communications**

By

<SXXXGPYY>

For

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<260910>

INTRODUCTION

The main objective of this **MCA01** is to model a double sideband suppressed carrier communications system (**DSB-SC**) for the communication of **speech** and **DTMF** signals in an **underwater medium**. The **maximum frequency content** of all these signals is assumed to be $F_V = 4000\text{Hz}$ and the “one-sided” **channel bandwidth** is said to be $B = 14000\text{Hz}$. A receiver **low-pass filter** must be designed to recover the signal $x_r(t) \approx s(t)$ (see **Fig. 3**).

The channel itself is modeled as a composition of two basic systems, an **ideal, linear phase, low-pass filter** T_C , with **cut-off frequency**, $f = F_L$, and a noise signal **summing system**, T_N , where the channel underwater acoustics **noise signal**, $n(t)$, is added to the output of the linear phase filter T_C . This DSB-SC communications channel is given the name $T_{CN} = T_{AM}$. Thus, it is given by cascading the sub-systems, T_C and T_N (see **Fig. 2**).

The overall sampling frequency of the DSB-SC communications system must be set to a value greater than **twice the maximum frequency** (**Nyquist-Shannon sampling theorem**) content of the output of the demodulator. That is, $F_S > 2(2f_c + F_V)$, where f_c is the **carrier frequency** of the modulator (demodulator) and must be $f_c = 8640$. The input signal $x_m(t)$ to the modulator is the sum of a “**wanted**” speech signal $s(t)$ and an “**unwanted**” **interference signal** $g(t)$. After the demodulator, an ideal, linear phase, low-pass filter, T_L is used, with cut-off frequency $F_M = 4000\text{Hz}$, to recover the **speech or voice signal** or **DTMF** signal $s(t)$ (see **Fig. 1**).

DESIGN PROJECT: Use [dtmfad](#) instead of [star_story](#)

For the successful completion of this assignment, each member in each group must study, run, and modify the provided m-script in the link [DSB-SC simulation](#) according to the specifications provided from **01** to **10**, below. Before starting with the assignment, each student must modify the provided m-script file by **displaying an adequate segment** of each signal in the time domain, in addition to the existing signals, for better visual interpretation.

01.- Access and review **all** hyperlinks provided.

02.- **Run** the given program, study **all** the graphs, and **observe** that the ideal, linear phase, [band-pass filter](#) **does eliminate** given interference signal $g(t)$.

$$g(t) = \cos(2\pi 4400t) + \cos(2\pi 4900t) + \cos(2\pi 5400t)$$

03.- **Verify** that **channel filter** is an “all pass” : $h_c(t) = \delta(t)$.

04.- **Verify** that **receiver’s band-pass filter** is: $h_R(t) = \delta(t)$.

05.- **Change** the carrier frequency to $f_c = 8640$.

06.- **Increase** value of **noise parameter factor** n_{ATT} by **40 dB**; that is, $\left(\frac{1}{\sqrt{24000}} \text{ to } \frac{1}{\sqrt{2.4}}\right)$ in order to go from about **–44 dB** to about **–4 dB SNR** and, thus, decreasing the **SNR** in the channel output $y_{co}(t)$ by about **40 dB**.

07.- A linear phase, [low-pass filter](#), T_L , with cut-off frequency $F_V = 4000\text{Hz}$, for the receiver was designed. A [figure of merit](#) for this assignment, is the **time-delay**, $T_d = \left(\frac{M_L - 1}{2}\right) T_s$, where M_L is the order of the filter and $T_s = \frac{1}{F_s}$ is the overall [sampling time](#) or [sampling period](#) of the system. For this work, in order to enhance [impulse response function](#), set the order of the low pass filter to an [odd value](#) of $M_L = N_{low} = 1323$.

08.- **Generate:** **1)** magnitude of the spectrum of the output signal $y_{co}(t)$ of the channel, **2)** the magnitude of the spectrum of the output signal $y_d(t)$ of the demodulator, **3)** the impulse response function $h_L(t)$ of the receiver’s low pass filter, **4)** the magnitude of its spectrum, $H_L(f)$, **5)** the receiver’s output signal $x_r(t)$, and **6)** the magnitude of its spectrum, $X_r(f)$.

09.- **Select** a word or message, composed of a [symbol sequence](#), and determine at the receiver the two **decimal digits** ([dits](#)) transmitted in this [digital communication](#).

10.- **Proceed** to calculate the information [baud rate](#) (D_r in symbols/second), [symbol-bit rate](#) (S_b in bits/symbol), the “info” [gross bit rate](#) (R_b in bits/second), and the [code rate](#) (C_r dimensionless), for [DTMF](#) digital communication operation.

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TAKE-HOME ASSIGNMENT GUIDELINES:

A.- This take-home work does not require a written result. It requires for each student group to send a **.zip** folder with the following items provided in the list below. There should be **NOTHING else** inside the requested **.zip** folder:

- i.- Original, INEL4301_SYYY_GPZZ_MCA02_260910.docx
- ii.- Original, INEL4301_SYYY_GPZZ_MCA02_260910pdf
- iii.- Original, unaltered, **dtmfad.wav** and **star_story.wav**
- iv.- Original, unaltered, **inel4301s000gp00mca02.m**
- v.- One modified **inel4301s000gp00sn00mca02.m** per student.

Refers to the class section number: **sxxx**

Refers to the student's group number: **gpyy**

B.- Name of **.zip** folder and the name of the **e-mail subject**:

INEL4301_SXXX_GPYY_MCA02

C.- Send e-mail, with the correct **e-mail subject**, and **.zip** file, with appropriate name, to the following e-mail address:

domingo.rodriquez1@upr.edu

STANDARD DEMERIT TABLE

01.- MATLAB m-script file does NOT execute	-05 pts.
02.- Correct script file NOT included	-05 pts.
03.- Missing INEL4301_SYYY_GPZZ_MAC01_260825.docx	-03 pts.
04.- Missing INEL4301_SYYY_GPZZ_MCA01_260825.pdf	-03 pts.
05.- Missing class section number sxxx	-02 pts.
06.- Unzipped folder does NOT have same name	-02 pts.
07.- Missing student's group number gpyy	-05 pts.
08.- Missing or incorrect e-mail subject	-05 pts.

Late works will not be accepted!!!

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```
%inel4301s000gp00sn00mca02
%Prof. Domingo Antonio Rodriguez
%Name of Student, sn01, DTMF_WORD, DITS, %XX of Contribution
%Name of Student, sn02, DTMF_WORD, DITS, %XX of Contribution
%Name of Student, sn03, DTMF_WORD, DITS, %XX of Contribution
%Name of Student, sn04, DTMF_WORD, DITS, %XX of Contribution
%Name of Student, sn05, DTMF_WORD, DITS, %XX of Contribution
%*****Prof. D. Rodriguez*****
%*****DSB-SC COMMUNICATIONS SYSTEMS AND GAUSSIAN NOISE*****
%*****ANALOG AND DIGITAL COMMUNICATIONS SIMULATIONS*****
%*****
clear all                                %Clear working space
close all
%*****
disp('*****')
disp('Begin Simulation for Underwater Communications.')
disp('*****')
%*****
%*****Wanted Signal Input/Output*****
%*****
[sig,Fs]=audioread('dtmfad.wav'); %Get overall sampling frequency
save dsig.txt sig -ascii           %Save signal as dsig.txt file
load -ascii dsig.txt;             %Load ascii file dsig.txt
wsiz=length(dsig);               %Get the length of the "dsig" signal
%*****
%*****Parameter Setting*****
%*****
Ts=1/Fs;                          %Sampling time or temporal resolution
Nq=wsiz;                          %Number of samples of wanted signal
Vq=Nq*Ts;                         %Observation window or signal duration
fincq=1/Vq;                       %Frequency resolution of wanted signal
fkq=-(Fs/2):fincq:(Fs/2)-fincq; %Frequency axis of wanted signal
tq=0:Ts:Vq-Ts;                   %Time axis of wanted signal
%
fV=4000;                          %Cut-off frequency of wanted signal
Wn=(2*fV/Fs);                    %Normalized cut-off frequency
Lp=(1/567);                      %Proportionality factor
Nh=2*floor(Lp*Nq)+1;             %Low-pass filter length
hL=fir1(Nh,Wn);                  %Impulse response of low-pass filter
dsigL=conv(dsig,hL);             %Filtered wanted signal up to fV=4000 Hz
%
f1=4400;                         %Input interference frequency
f2=4900;                         %Input interference frequency
f3=5400;                         %Input interference frequency
g1=cos(2*pi*f1*tq);              %Unwanted input signal
g2=cos(2*pi*f2*tq);              %Unwanted input signal
g3=cos(2*pi*f3*tq);              %Unwanted input signal
g=(1/80)*g1+(1/160)*g2+(1/240)*g3; %Sum of unwanted signals
gmax=max(g);                     %Maximum value of sum
g=transpose((1/gmax)*g);         %Normalized interference signal
%*****
%*****SIGNALS AND SYSTEMS MODELING*****
fc=9900;                         %Carrier frequency
c=transpose((cos(2*pi*fc*tq))); %Carrier signal
wsig=dsigL(1:Nq,1);             %Wanted input signal
xm=wsig+g;                       %Sum of wanted and unwanted signals
xc=xm.*c;                       %Modulated wanted signal plus interference
%*****
%*****BAND-PASS FILTER*****
Nb=1001;                        %Order of band-pass filter
```

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Wb=[(2*4640)/Fs (2*12640)/Fs]; %Vector of normilized cut-off freqs.
hB=fir1(Nb-1,Wb,'bandpass'); %Impulse response of band-pass filter
%*****CHANNEL INPUT SIGNAL*****
yci=conv(xc,hB); %Removal of unwanted interferences
Nyci=length(yci); %Order of channel input signal
tyci=0:Ts:(Nyci-1)*Ts; %Time axis for yci(t)
fryci=1/(Nyci*Ts); %Frequency resolution for Yci(f)
fayci=-(Fs/2):fryci:(Fs/2)-fryci; %Frequency axis of wanted signal
%*****CHANNEL NOISE*****
nsig=randn(size(yci)); %Generation of AWGN signal: noise signal
nnsig=(1/max(abs(nsig)))*nsig; %Normalized noise signal
nATT=sqrt(1/24000); %Noise attenuation about -44 dB SNR
nsig=nATT*nnsig; %Attenuated noise present in the channel
%*****CHANNEL OUTPUT*****
yco=yci+nsig; %Channel output signal with Gaussian noise
tqn=0:Ts:(length(yco)-1)*Ts; %New time axis for output signal
cn=transpose((cos(2*pi*fc*tqn))); %New carrier signal
yd=yco.*(2*cn); %Demodulation stage
%*****LOW-PASS FILTER*****
Nlow=663; %Designed low-pass filter order
Wlow=(2*fV)/Fs; %Normalized cut-off frequency
hlow=fir1(Nlow-1,Wlow); %FIR low-pass impulse response function
taxis_hlow=0:Ts:(length(hlow)-1)*Ts;
fres_hlow=1/((length(hlow))*Ts);
faxis_hlow=(-Fs/2):fres_hlow:(+Fs/2)-fres_hlow;
%*****RECEIVER OUTPUT*****
xr=conv(yd,hlow); %Low pass filtering at the receiver
taxis_xr=0:Ts:(length(xr)-1)*Ts;
fres_xr=1/((length(xr))*Ts);
faxis_xr=(-Fs/2):fres_xr:(+Fs/2)-fres_xr;
%*****SPECTRA*****
fwsig=fft(wsig); %Spectrum of wanted signal (Fund. Reg.)
sfwsig=fftshift(fwsig); %Spectrum of wanted signal (Princ. Reg.)
asfwsig=abs(sfwsig); %Magnitude of spectrum of wanted signal
%
fxc=fft(xc); %Spectrum of modd. signal (Fund. Reg.)
sfxc=fftshift(fxc); %Spectrum of modd. signal (Princ. Reg.)
asfxc=abs(sfxc); %Magnitude of spectrum of modd. signal
%
fyci=fft(yci); %Spectrum of input signal (Fund. Reg.)
sfyci=fftshift(fyci); %Spectrum of input signal (Princ. Reg.)
asfyci=abs(sfyci); %Magnitude of spectrum of input signal
%
fyco=fft(yco); %Spectrum of output signal (Fund. Reg.)
sfyco=fftshift(fyco); %Spectrum of output signal (Princ. Reg.)
asfyco=abs(sfyco); %Magnitude of spectrum of output signal
%
fyd=fft(yd); %Spectrum of output signal (Fund. Reg.)
sfyd=fftshift(fyd); %Spectrum of output signal (Princ. Reg.)
asfyd=abs(sfyd); %Magnitude of spectrum of output signal
%
fnsig=fft(nsig); %Spectrum of noise signal (Fund. Reg.)
sfnsig=fftshift(fnsig); %Spectrum of noise signal (Princ. Reg.)
asfnsig=abs(sfnsig); %Magnitude of spectrum of noise signal
```

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```
%
fhlow=fft(hlow); %Spectrum of IRF (Fund. Region)
sfhlow=fftshift(fhlow); %Spectrum of IRF (Princ. Region)
asfhlow=abs(sfhlow); %Magnitude of spectrum IRF
%
fxr=fft(xr); %Spectrum of xr(t) (Fund. Region)
sfxr=fftshift(fxr); %Spectrum of xr(t) (Princ. Region)
asfxr=abs(sfxr); %Magnitude of spectrum IRF
%*****PLOTS*****
plot(tq,wsig);
axis([0 (length(wsig)*Ts) min(wsig) max(wsig)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Wanted signal s(t) in the Time Domain')
grid
%
figure
plot(tq,g);
axis([0 (length(g)*Ts) min(g) max(g)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Interference signal g(t) in the Time Domain')
grid
%
figure
plot(tq,xm);
axis([0 (length(xm)*Ts) min(xm) max(xm)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Modulating signal xm(t) in the Time Domain')
grid
%
figure
plot(tq,xc)
axis([0 (length(xc)*Ts) min(xc) max(xc)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Modulated Signal xc(t)')
grid
%
figure
plot(ty,nn,nnsig)
axis([0 (length(nnsig)*Ts) min(nnsig) max(nnsig)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Normalized Noise Signal nn(t)')
grid
%
figure
plot(ty,yci)
axis([0 (length(yci)*Ts) min(yci) max(yci)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Channel Input Signal yci(t)')
grid
%
figure
plot(ty,yco)
axis([0 (length(yco)*Ts) min(yco) max(yco)]);
xlabel('Time in Seconds')
```

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```
ylabel('Amplitude')
title('Channel Output Signal yco(t)')
grid
%
figure
plot(fkq,asfwsig)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum S(f) of Wanted Signal s(t)')
grid
%
figure
plot(fkq,asfxc)
xlabel('Freq in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Xc(f) of Modulated Signal xc(t)')
grid
%
figure
plot(fayci,asfyci)
xlabel('Freq in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yci(f) of Channel Input Signal yci(t)')
grid
%
figure
plot(fayci,asfyco)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yco(f) of Channel Output Signal yco(t)')
grid
%
figure
plot(fayci,asfyd)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum Yd(f) of Demodulator Output yd(t)')
grid
%
figure
plot(taxis_hlow,hlow);
axis([0 (length(hlow)*Ts) min(hlow) max(hlow)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Impulse Response Function of Receiver's Low-Pass Filter')
grid
%
figure
plot(faxis_hlow,asfhlow)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum of Impulse Response Function hlow(t)')
grid
%
figure
plot(taxis_xr,xr);
axis([0 (length(xr)*Ts) min(xr) max(xr)]);
xlabel('Time in Seconds')
ylabel('Amplitude')
title('Receiver's Low-Pass Filter Output Signal xr(t)')
```

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```
grid
%
figure
plot(faxis_xr,asf_xr)
xlabel('Frequency in Hertz')
ylabel('Magnitude')
title('Magnitude of Spectrum of Receiver''s Output x_r(t)')
grid
%
%*****OUTPUTS*****
sound(wsig,Fs) %Sound of Wanted Signal
disp('Sound of Wanted Signal')
pause(20)
sound(g,Fs) %Sound of Interference Signal
disp('Sound of Interference Signal')
pause(20)
sound(xm,Fs) %Sound of Modulating Signal
disp('Sound of Modulating Signal')
pause(20)
sound(xc,Fs) %Sound of Modulated Signal
disp('Sound of Modulated Signal')
pause(20)
sound(yci,Fs) %Sound of Channel's Input Signal
disp('Sound of Channel''s Input Signal')
pause(20)
sound(yco,Fs) %Sound of Channel's Output Signal
disp('Sound of Channel''s Output Signal')
pause(20)
sound(xr,Fs) %Sound of Channel's Output Signal
disp('Sound of Receiver''s Output Signal')
%*****
disp('*****')
disp('Simulation ended succesfully.')
disp('*****')
```